Memory map and peripheral registers

Data address = 8 bits (Rs of LD/ST). addr_decode: is_ram = ~a[7]; is_per = a[7]&a[6]&a[5]&a[4]; a 4-to-16 decoder on a[3:0] enabled by is_per selects one peripheral register. A load returns OR over (value AND select), so every unselected or unimplemented address reads 0x00. A write needs EX/MEM.v & memw & the select line, and it happens on the clock edge that ends MEM. RAM writes also need is_ram, so 0x80-0xEF can never alias into the 7-bit RAM.

Address	Name	Access	Reset	Bit fields / behaviour
0x00-0x7F	DATA RAM	R/W	undefined (0 in sim)	128 x 8 built-in RAM, async read, write on rising edge (WE = v&memw&is_ram)
0x80-0xEF	reserved	-	-	reads 0x00, writes ignored
0xF0	IN_STAT	R	live	[1:0] MODE, [2] START, [3] DOOR_CLOSED, [4] TRIP (readback), [5] MOTOR_ACTUAL (readback), [7:6] 0
0xF1	WATER_LEVEL	R	live	[7:0] unsigned water level
0xF2	TEMP	R	live	[7:0] temperature readback (no regulation in core scope)
0xF3	DRUM_RPM	R	live	[7:0] drum speed readback
0xF4	ACT (actuator latch)	R/W	0x00	[0] VALVE_IN [1] HEATER [2] MOTOR_EN [3] MOTOR_DIR [4] PUMP_OUT [5] DOOR_LOCK [6] BUZZER [7] reserved (written as 0, reads 0)
0xF5	DISPLAY	R/W	0x00	[7:0] status code driven to DISPLAY pins
0xF6	TIMER_COUNT	R	0x00	current count; writes ignored
0xF7	TIMER_LIMIT	R/W	0x00	compare value; 0 disables comparison; writing it never sets MATCH
0xF8	TIMER_CTRL	R/W	0x00	[0] EN (stored, reads back); [1] CLR: write 1 = clear COUNT and MATCH at that edge (self-clearing, reads 0); [7:2] ignore / read 0
0xF9	TIMER_STATUS	R	0x00	[0] MATCH (sticky); writes ignored (clear through TIMER_CTRL.CLR)
0xFA-0xFF	unimplemented	-	-	reads 0x00, writes ignored

Physical outputs (main circuit)

VALVE_IN, HEATER, MOTOR_EN_REQ (commanded), MOTOR_DIR, PUMP_OUT, DOOR_LOCK, BUZZER = ACT bits; DISPLAY[7:0]; MOTOR_ACTUAL = MOTOR_EN_REQ & DOOR_CLOSED & ~TRIP (safety-gated physical enable); TRIP; HALTED; plus debug pins PC, PC_ID/EX/MEM/WB, V_ID/EX/MEM/WB, STALL, FLUSH, FWD_A/B (00 RF, 01 MEM/WB, 10 EX/MEM), RF_WE/WA/WD, ST_COMMIT/ADDR/DATA, TMR_COUNT, TMR_MATCH.

Timer behaviour (timer8)

Registers COUNT, LIMIT, EN, MATCH on the global clock, not stalled and not affected by HALT. At every rising edge: if clr (valid store to 0xF8 with data bit 1) then COUNT <- 0, MATCH <- 0 (priority over increment and compare); else if EN then COUNT <- COUNT+1 (inc8, mod 256) and MATCH <- MATCH | (COUNT+1 == LIMIT & LIMIT != 0) (eq8, zero8). EN <- data bit 0 on any CTRL write; it takes effect from the next edge. Typical use: write LIMIT, then CTRL = 0x03 (clear + enable): COUNT reads 1 after the first following edge, and MATCH rises together with COUNT == LIMIT. CTRL = 0x02 stops and clears; 0x00 pauses; 0x01 resumes. Reset: EN=0, COUNT=LIMIT=MATCH=0.

Data RAM use by program.asm

0x10+m target level, 0x14+m wash ticks, 0x18+m spin ticks, 0x1C+m display base (m = MODE 0..3); all written by the program before use.